AI Resume Analyzer & Job Recommendation System

End-to-End Workflow & Architectural Specification

1. Ingestion & Privacy

- In-Memory PDF (pypdf)
- DOCX Parser (python-docx)
- Zero Persistent Storage
- Anti-Bias Data Scrubber
- Section Segmenter

2. Feature Extraction

- 60 Skills (7 Domains)
- Boundary Regex Matcher
- Multi-Word Lookahead
- Degree & Graduation Year
- Experience Years Estimator

3. Hybrid Matching

- TfidfVectorizer (ngram 1,2)
- Sublinear TF Scaling
- Contextual Cosine Sim
- 60% Skill + 30% TFIDF
- Top 3 Role Recommendations

4. Outputs & Roadmap

- Skill Gap Matrix (Matched/Missing)
- 3-Phase Upskilling Roadmap
- Curated Free Courses & Docs
- 3 Capstone Projects (STAR)
- ReportLab PDF & Markdown Export

System Matching Function Formulation:

*Match Score = 0.60 * Skill_Coverage_Pct + 0.30 * TFIDF_Cosine_Pct + 0.10 * Preferred_Bonus_Pct*

Evaluated across 10 benchmark enterprise job roles and 60 technical skills. Demographic variables strictly excluded.